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DEEP LEARNING RIPARIAN LAND COVER CLASSIFICATION METHODOLOGY USING AERIAL PHOTOGRAPH

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Abstract

Land cover classification has been a vital component of hydraulic research. Due to the global climate condition changes, massive flooding events have caused researchers to re-evaluate the flow capacity of various rivers. In response to the challenges of land cover classifications, many researchers have started experimenting with AI-based methods. As we know, the ortho aerial photograph is an accurate representation of Earth's surface. Accordingly, this paper presents an example of how an ortho aerial photograph can be used to classify the land cover in the river area. For this study, the ortho aerial photograph was separated into three parts: one for train, one for test, and another part for valid. The index values of the classifications for different vegetation categories were computed using our algorithm. To conclude, this DeepLabV3+ model can provide almost 0.9 in overall accuracy and Macro-F1, respectively.

Keywords: Riparian land-cover classification; Deep learning; Semantic segmentation

1. INTRODUCTION

In early July 2018, a large flood of approximately $4,500 \text{ m}^3\text{s}^{-1}$ occurred in the lower Asahi River in Okayama Prefecture, which was the target survey site. The surveyed river reached record water levels due to riparian vegetation, such as dense trees and bamboo forests (Yoshida *et al.*, 2020). Therefore, proper LCC mapping is required in flood modeling to estimate flow resistance parameters due to riparian vegetation. In view of these issues, this study investigates the proposed deep learning (DL)-based approach to map LCC in riparian areas using only RGB. However, the new land cover classifications (LCC) mapping method proposed here outperforms previous unsupervised methods (clustering) in distinguishing the most dominant riparian vegetation species in the region (*i.e.*, woody vegetation and bamboo forest) (Yoshida *et al.*, 2022). Finally, the proposed LCC reasonably estimates the hydrodynamic roughness of the spatial distribution in the 2018 Asahikawa flood model (Yoshida *et al.*, 2022). Overall, this study is expected to help policymakers develop balanced scenarios for flood control and ecosystem management tasks, taking into account waterside LCCs.

2. TEXTS

Figure 1 (left) depicts our study site, which is located on the lower Asahi River, a Class I (state-controlled) river in Japan, flowing through Okayama prefecture into the Seto Inland Sea. The targeted domain of this study is from 13.2-17.4KP, as shown in Figure 1 (left). We have accumulated a certain amount of fieldwork photos, which has provided us with a good foundation for drawing the relevant true label (TL). Since making photos of multiple seasons over many years, the differences can be clearly observed, especially the changes in color and texture feature. Based on the above situation, this study used DL technology in the semantic segmentation challenge to infer LCC. Regarding DL method, TL mapping is a very important input data. TL mapping is mainly based on aerial photographs, and supplemented with years of field observation experience. As Figure 2, because it was manually drawn, time and effort are necessary in the early stage, subsequently, TL mapping can obtain a relatively high degree of confidence due to the check based on the field observations. Due to our limited input data and considering the ratio of train, valid, and test datasets (8:1:1) and the ratio of each label in each dataset (1:1:1), the March 2017 aerial photographs were segmented as Figure 3. The train part as Figure 4 was used as input data [Input] to be trained by the [DeepLabV3 model], and the trained file was determined as [Trained model (RGB)]. At that juncture, the test part was inferred in [Trained model (RGB)]

and get the prediction result, and finally the inferred LCC mapping will be compared with TL by confusion matrix. To conclude, the operation steps are following as, train part: [Input from train part] → [DeepLabV3 model] → [Trained model (RGB)], inference part: [Input from test part] → [Trained model (RGB)] → [Inference] + [TL] → [Confusion matrix].

Regarding the confusion matrix of this study, overall accuracy (OA) and Macro-F1 were chosen to be used, considering the comparative indices to examine the overall and individual label performance. The meaning of each part in the confusion matrix is shown in Table 1, and a sample of confusion matrix is made as shown in Table 2. In conclusion, Table 3 shows the confusion matrix of the RGB-based March 2017 results.

3. FIGURES AND TABLES

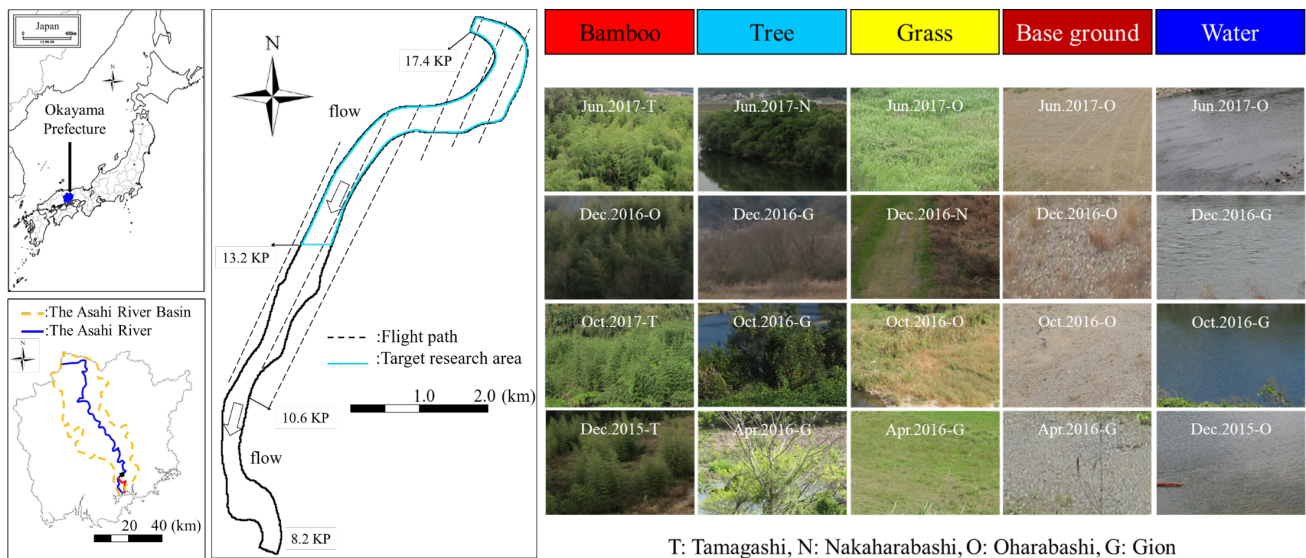


Figure 1 Location of the Asahi River in Japan with the kilometer post (KP) values representing the longitudinal distance (km) from the river mouth (left), and field observation results (right)

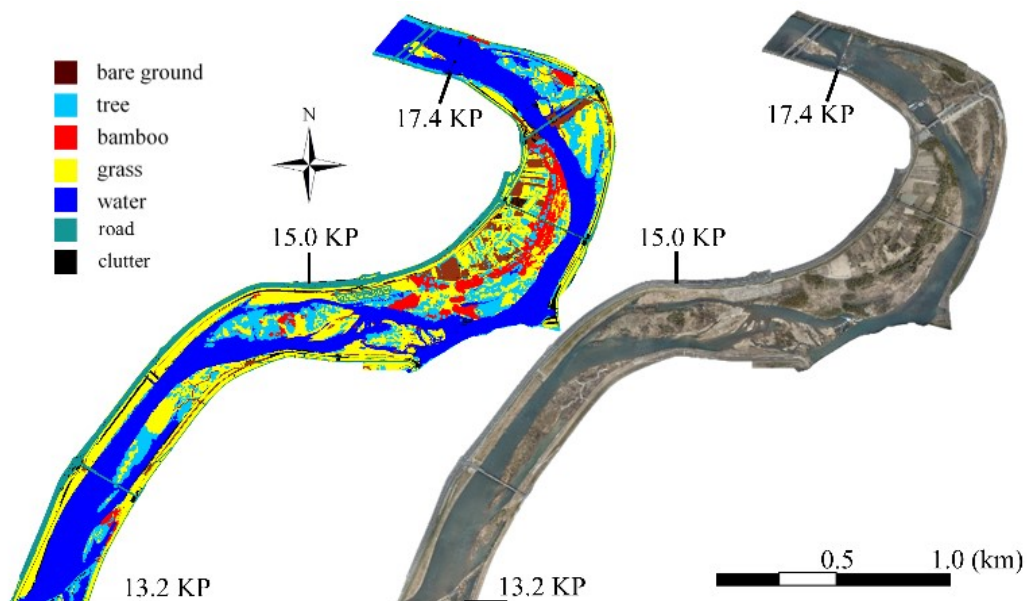


Figure 2 Sample of true label mapping and ortho-aerial photograph of the targeted area in March, 2017.

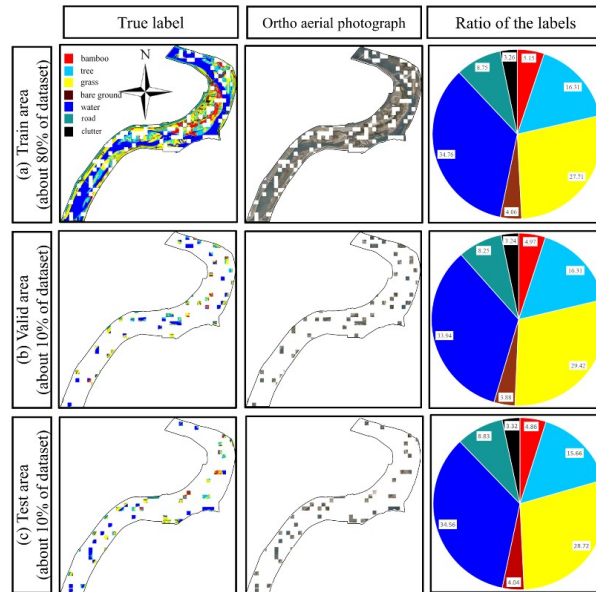


Figure 3 Spatial distribution of the March, 2017 dataset in the (a) train, (b) valid, and (c) test areas.

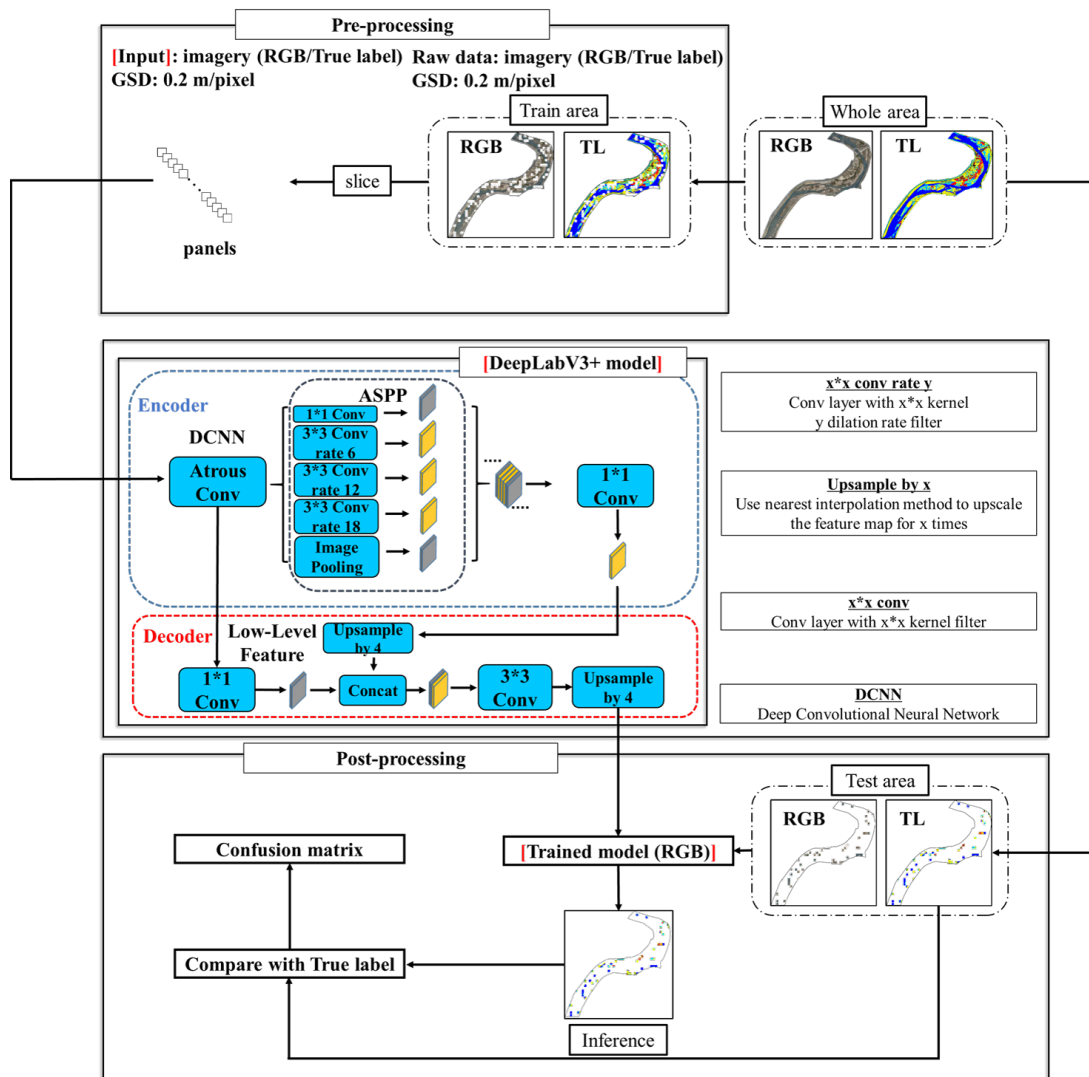


Figure 4 Processing of mapping LCC with DeepLabV3+ Model

Table 1. Confusion matrix valuation index for Accuracy

Confusion matrix valuation index		
Symbol	Definition	Formula
Precision (X)	The ratio of the pixels for correctly predicted as X to all the pixels predicted as X ¹⁾	$TP-X^2) / PR-X^3)$
Recall (X)	The ratio of the pixels for correctly predicted as X to all the pixels true label as X	$TP-X / TL-X^4)$
F1-score (X)	F1-score is the weighted average of Precision and Recall.	$2 * Precision (X) * Recall (X) / (Precision (X) + Recall (X))$
OA	Overall accuracy value of the confusion matrix	$\sum TP-X / \text{Amount of total pixels}$
Macro-F1	Macro-F1 is the average of all F1-score.	$\sum F1\text{-score (X)}/\text{amounts of labels}$

¹⁾X or Y includes 5 labels (B: Bamboo, T: Tree, G: Grass, BG: Bare Ground, W: Water);

²⁾TP-X is the amount of the pixels that true label and prediction are all X;

³⁾PR-X is the amount of the pixels that prediction is X;

⁴⁾TL-X is the amount of the pixels that true label is X.

Table 2. Sample of Confusion matrix

Sample of Confusion matrix							
	B	T	G	BG	W	Total	Recall (%)
B	TP-B	E-B/T	E-B/G	E-B/BG	E-B/W	TL-B	TP-B/TL-B
T	E-T/B	TP-T	E-T/G	E-T/BG	E-T/W	TL-T	TP-T/TL-T
G	E-G/B	E-G/T	TP-G	E-G/BG	E-G/W	TL-G	TP-G/TL-G
BG	E-BG/B	E-BG/T	E-BG/G	TP-BG	E-BG/W	TL-BG	TP-BG/TL-BG
W	E-W/B	E-W/T	E-W/G	E-W/BG	TP-W	TL-W	TP-W/TL-W
Total	PR-B	PR-T	PR-G	PR-BG	PR-W	Total pixels	
Precision (%)	TP-B/PR-B	TP-B/PR-T	TP-B/PR-G	TP-B/PR-BG	TP-B/PR-W		

OA, Macro-F1

E-X / Y: Amount of the pixels where true label is X, prediction is Y.

Table 3. 2 m resolution test area RGB-based result

2 m resolution test area RGB-based result									
	B	T	G	BG	W	R	C	Total	Recall (%)
B	199972	21107	12728	258	0	61	92	234218	85.38
T	27745	582370	123393	2693	20971	861	9411	767444	75.88
G	19632	128550	1262665	31170	4366	8958	28725	1484066	85.08
BG	560	367	46527	178830	1775	207	1621	229887	77.79
W	880	19500	4705	2336	1723609	858	7636	1759524	97.96
R	0	189	5989	2741	468	447099	7711	464197	96.32
C	231	3726	19369	2784	4022	4858	119278	154268	77.32
Total	249020	755809	1475376	220812	1755211	462902	174474	5093604	
Precision (%)	80.30	77.05	85.58	80.99	98.20	96.59	68.36		

OA = 0.91, Macro-F1 = 0.88

4. CONCLUSIONS

This study has proved that the DeepLabV3+ model can segment the riverine LCC just using RGB dataset from same period. Alternatively, The TL mapping also need more effective approach to produce, accordingly this will be the next step of our research. The test dataset from multi-periods and different places are on the way.

5. REFERENCES

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